

OpenNotebook - Project Scope & Milestones

Project Name: OpenNotebook

Project Type: AI Research Assistant Platform

Objective: Build a production-grade NotebookLM alternative with unlimited document ingestion, persistent conversational memory, advanced RAG pipelines, podcast generation, and collaborative AI workspaces.

Core Objectives

- Create a scalable AI research platform.
- Support multi-format document ingestion.
- Build persistent conversational memory.
- Implement hybrid retrieval pipelines.
- Generate grounded AI responses with citations.
- Support podcast and knowledge graph generation.
- Enable collaborative notebook-based workflows.
- Provide intelligent conversational interaction between users and AI using contextual memory, real-time retrieval, and streaming responses.

Technology Stack

Frontend: Next.js 15, React, TypeScript, TailwindCSS, shadcn/ui

Backend: FastAPI, LangGraph

Database: PostgreSQL

Vector Database: Qdrant

Queue System: Redis + Celery

AI Providers: Groq + Ollama fallback

Embeddings: BGE-small / all-MiniLM

STT: Whisper

TTS: Piper TTS / Coqui TTS

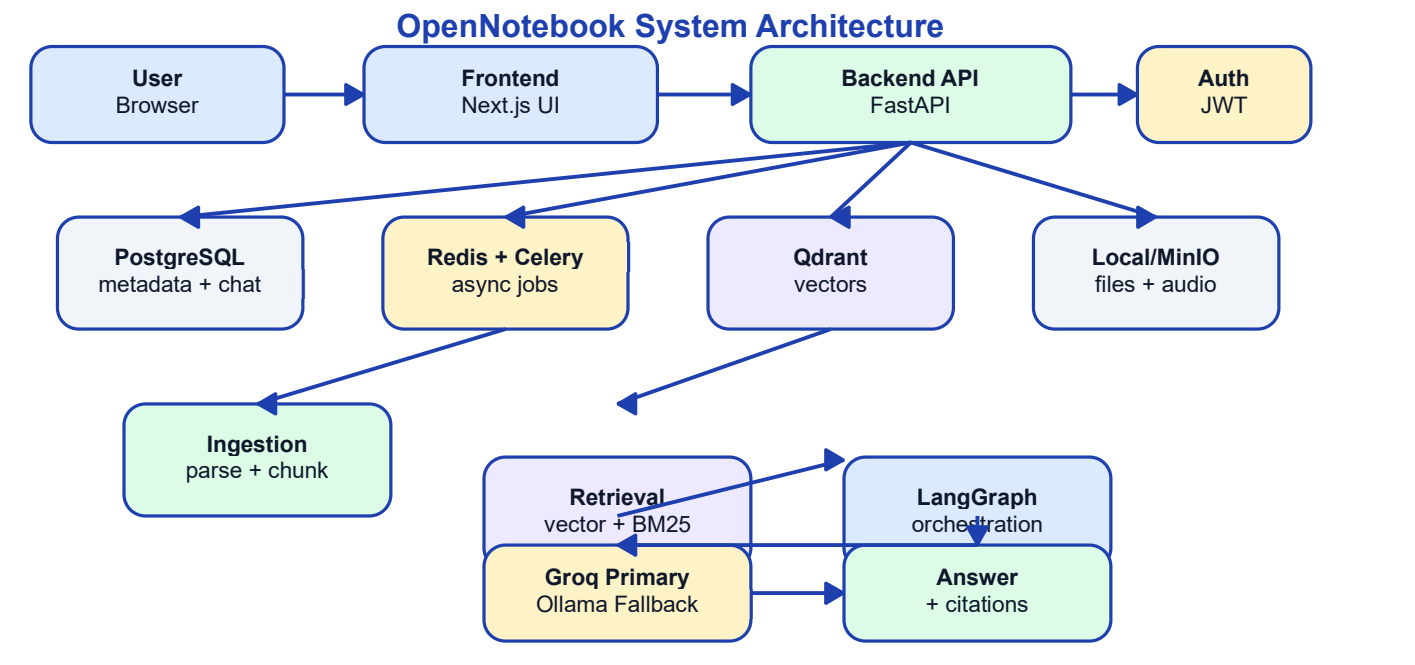
Deployment: Docker Compose

Major Features

- Unlimited notebook creation
- PDF, DOCX, PPTX, website, YouTube, and audio ingestion
- Advanced conversational RAG system
- Hybrid vector + BM25 retrieval
- Persistent memory system
- Podcast generation workflow
- Citation-based grounded answers
- Team collaboration and export support

System Architecture

Frontend communicates with FastAPI backend through REST APIs and streaming responses. Documents are processed asynchronously using Celery workers. Embeddings are stored in Qdrant, metadata in PostgreSQL, and orchestration handled using LangGraph. Groq acts as the primary LLM provider with Ollama local models as fallback.



Project Milestones

Phase	Milestone	Deliverables	Schedule
Phase 1	Infrastructure Setup	Docker, PostgreSQL, Redis, Qdrant setup	18-05-2026
Phase 2	Frontend Development	Modern NotebookLM-style UI	19-05-2026 to 20-05-2026
Phase 3	Authentication & Workspaces	JWT auth, workspace and notebook management	21-05-2026
Phase 4	Document Ingestion Pipeline	Upload, parsing, chunking, embeddings	22-05-2026 to 23-05-2026
Phase 5	RAG Chat System	Hybrid retrieval, citations, streaming chat	25-05-2026 to 26-05-2026
Phase 6	Memory Architecture	Short-term, summary, and vector memory	27-05-2026
Phase 7	Podcast Generation	AI script generation and TTS workflow	28-05-2026
Phase 8	Export & Analytics	PDF/DOCX export, analytics dashboard	29-05-2026
Phase 9	Testing & Optimization	Bug fixing, optimization, deployment	30-05-2026 to 01-06-2026

Conclusion:

OpenNotebook is designed as a scalable, modular, enterprise-grade AI research platform capable of handling massive document workloads and advanced AI-driven knowledge workflows.